This article is an outline for the checkout and start-up of Gas Turbine-Generator excitation systems.

1. Following is a description of the types of excitation systems generally encountered.
   a. SCT-PPT compound exciter.
   b. Potential Source or Bus Fed exciter.
   c. Shunt SCR exciter.
   d. Brushless exciter.
   e. Other Equipment Manufacturers (OEM) exciters.
   f. Digital exciters (the future).
2. Information required for installation, adjustments, checkout, and start-up.
   a. Turbine-Generator instruction manuals.
   b. Drawings
      - Customer drawings.
      - GE drawings.
   c. Generator information and data.
3. Test equipment required for adjustments and checkout.
4. Exciter system installation, adjustments, checkout, and start-up.
   a. Installation of the exciter equipment.
   b. Verification of wiring connections and control logic operation.
   c. Pre-start-up checks and adjustments.
   d. Generator start-up.

5. Other equipment associated with the exciter installation, checkout, and start-up.
    a. Protective relays.
    b. Generator controls and metering.
    c. Switchgear.
    d. Generator.
    e. Bus and cables.
6. Safety precautions during checkout and start-up.

Over the years, many types of excitation systems have been utilized. The following is a list of systems that are likely to be encountered during installation of a Gas Turbine-Generator. No attempt is made to describe the system's operation. For operational theory, refer to the appropriate instruction book for the system of concern.

A. SCT-PPT (Saturating Current Transformer-Power Potential Transformer), compound exciter.

B. Potential Source (sometimes referred to as Bus Fed) exciter.

C. Shunt SCR (Silicon Controlled Rectifier) compound exciter.

D. Brushless exciter.

E. Exciters constructed by Other Equipment Manufacturers (OEM's).

F. Digital exciters (the exciters that will begin appearing in the near future).

Missing from the above list, and not discussed in this article, is the rotating direct current exciter, (DC generator), that has been employed since the earliest days of power generation. The development of the silicon diode and silicon controlled rectifier has all but eliminated the use of rotating DC exciters (except those in existing installations). And a number of customers are retrofitting existing units to eliminate rotating DC exciters.

During the equipment planning stages, the customer is given an opportunity to select the type of exciter to be used, or to present requirements that may dictate the choice of exciters. The following are what may be expected to arrive at a jobsite, based on equipment selection.

The SCT-PPT compound system, Figure 1, has been in service, for many years on both steam and Gas Turbine-Generators. It has a reputation of being the most reliable system ever built. It has been replaced by the other systems, described below, but may still be furnished on request, and for this reason is mentioned here. In terms of cost it is probably on par with the Shunt SCR exciter (the most expensive). The major components consist of a saturating current transformer

(SCT) connected in the generator's neutral circuit; a power potential transformer (PPT), connected to the generators output; a power bridge rectifier (with linear reactors); and a voltage regulator with excitation controls. The SCT allows this exciter to sustain excitation on sudden load application or on system faults.

The Potential Source (or Bus Fed) excitation system, Figure 2, is currently being furnished on numerous installations, both steam and Gas Turbine-Generators. This exciter is the least expensive of the GE manufactured exciters used on Gas Turbine-Generators. The major components consist of a power potential transformer (PPT) connected in the generators output; a power bridge rectifier; and voltage regulator including excitation controls. This type of exciter has the disadvantage of not being able to maintain excitation during extended faults, nor being able to quickly lower excitation through an inverting bridge rectifier.

The Shunt SCR compound system, Figure 3, is currently being furnished on installations both steam and Gas Turbine-Generators. It is probably the most expensive system presently being used. The major components consist of a power current transformer (PCT) connected in the generators neutral circuit; a power potential transformer (PPT) connected to the generators output; a power bridge rectifier (with linear reactors); and a voltage regulator with excitation controls. The PCT allows this exciter to sustain excitation on sudden load application or on system faults.

The Brushless Excitation System, Figure 4, is presently being furnished on Gas Turbine-Generator installations. Its price is between the Potential Source and the Shunt SCR. The major components are: a rotating brushless exciter, a rotating power bridge rectifier "wheel", and a voltage regulator cabinet with excitation controls. The rotating exciter enables this system to sustain excitation for a brief period on sudden load application or on system faults. It also eliminates brushes and collector rings. Typically, the rotating equipment is manufactured by Electric-Machinery, and the voltage regulator may be manufactured by GE. In terms of physical space requirements, it is probably the most compact system.

Other equipment manufacturers (OEM's) may furnish excitation systems that would be used with GE Gas Turbine-Generators. There is a tendency to use GE equipment, but that may not always be the case.

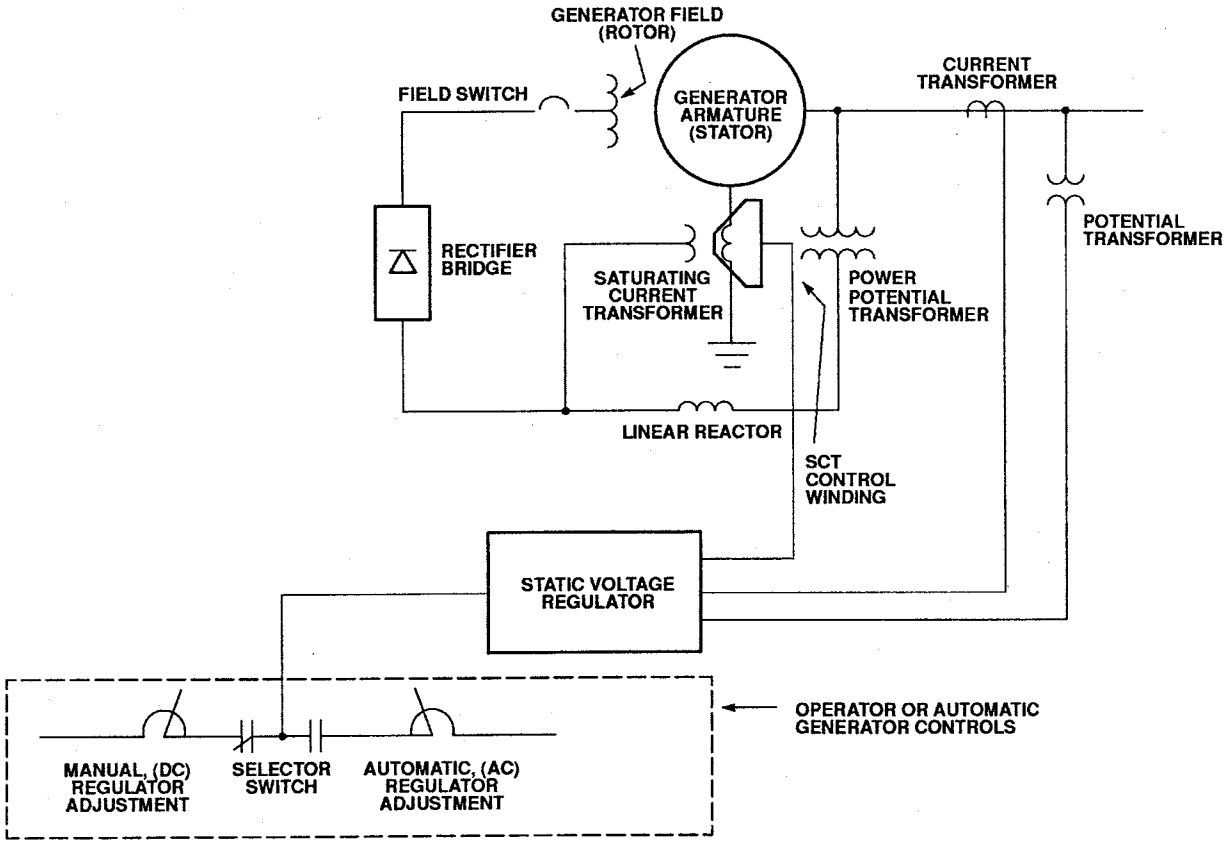


Figure 1. SCT – PPT Compound Exciter - Typical Layout

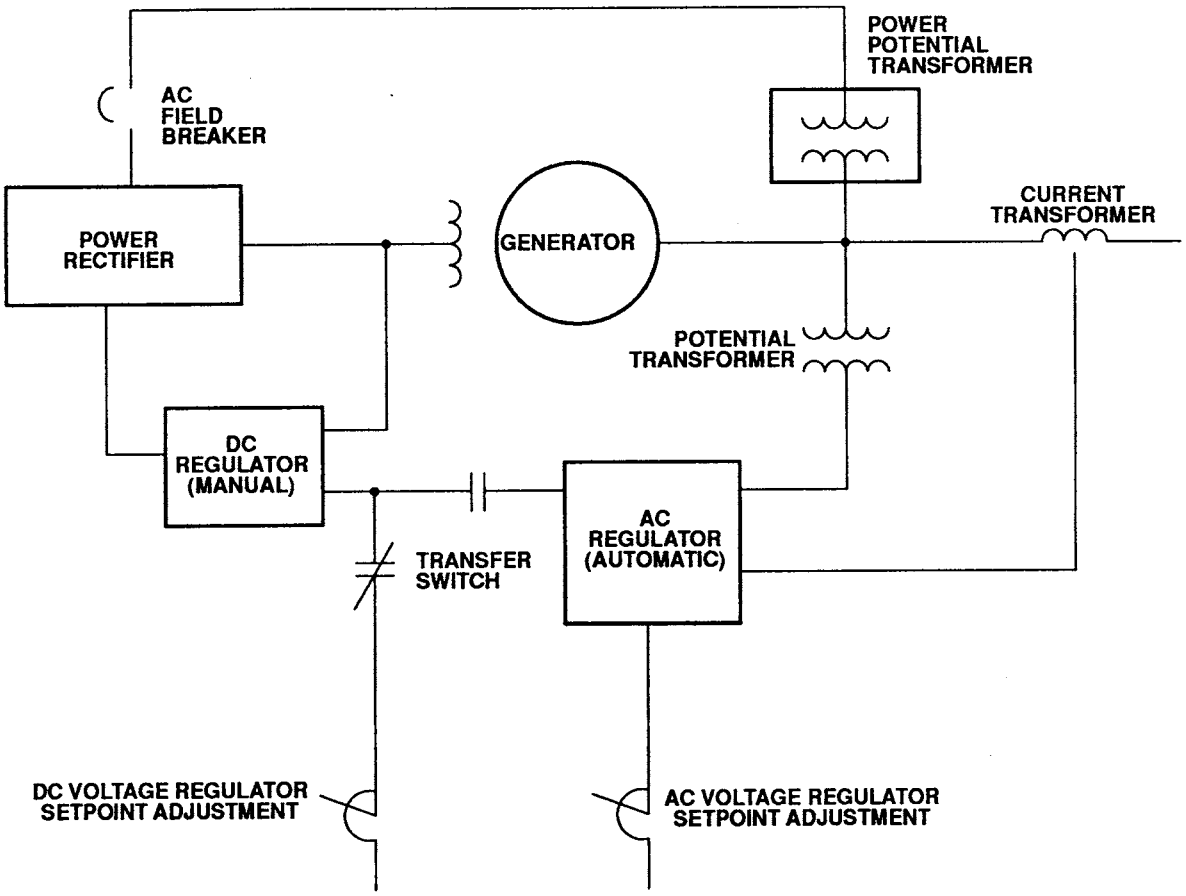


Figure 2. Potential Source Excitation System - Typical Layout

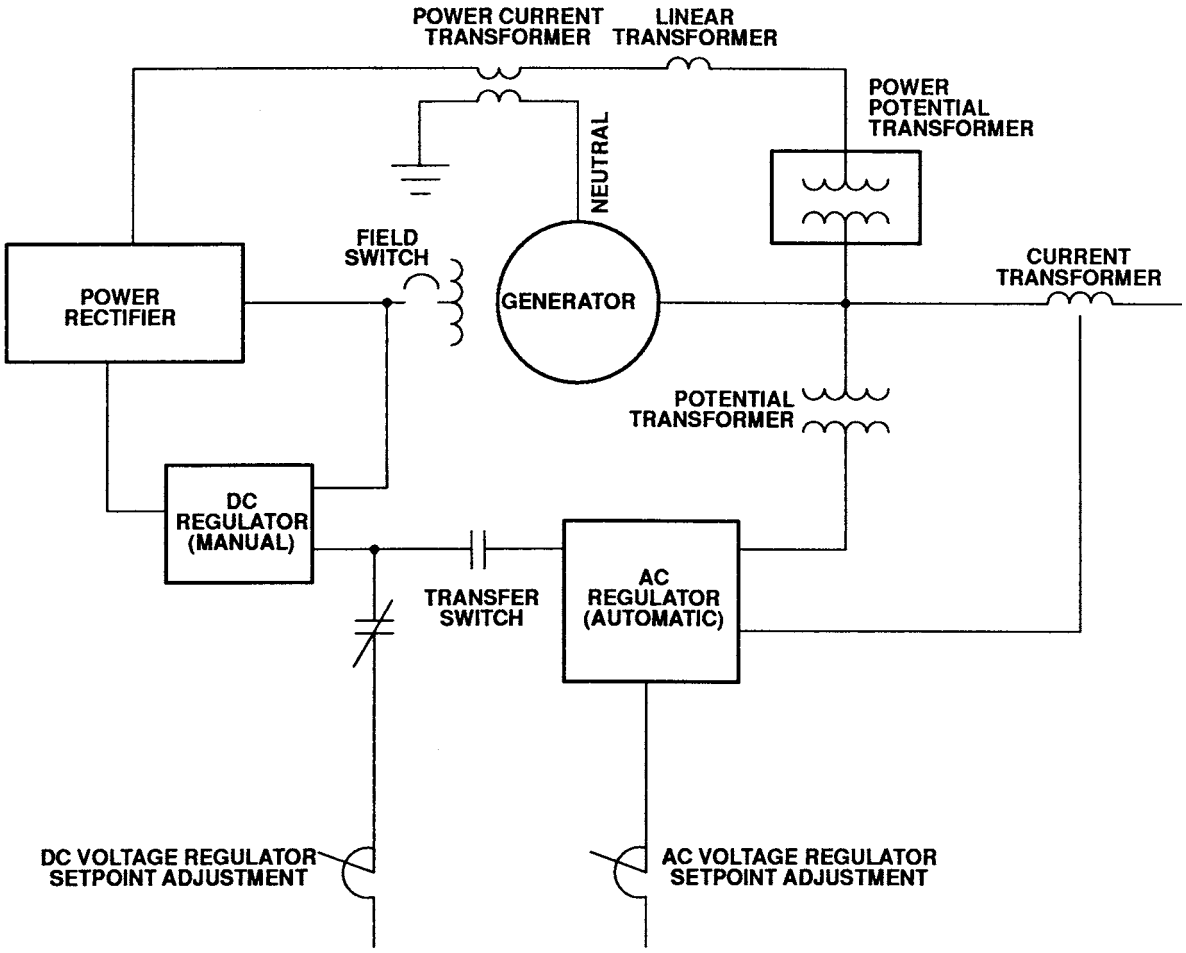


Figure 3. Shunt SCR Excitation System - Typical Layout

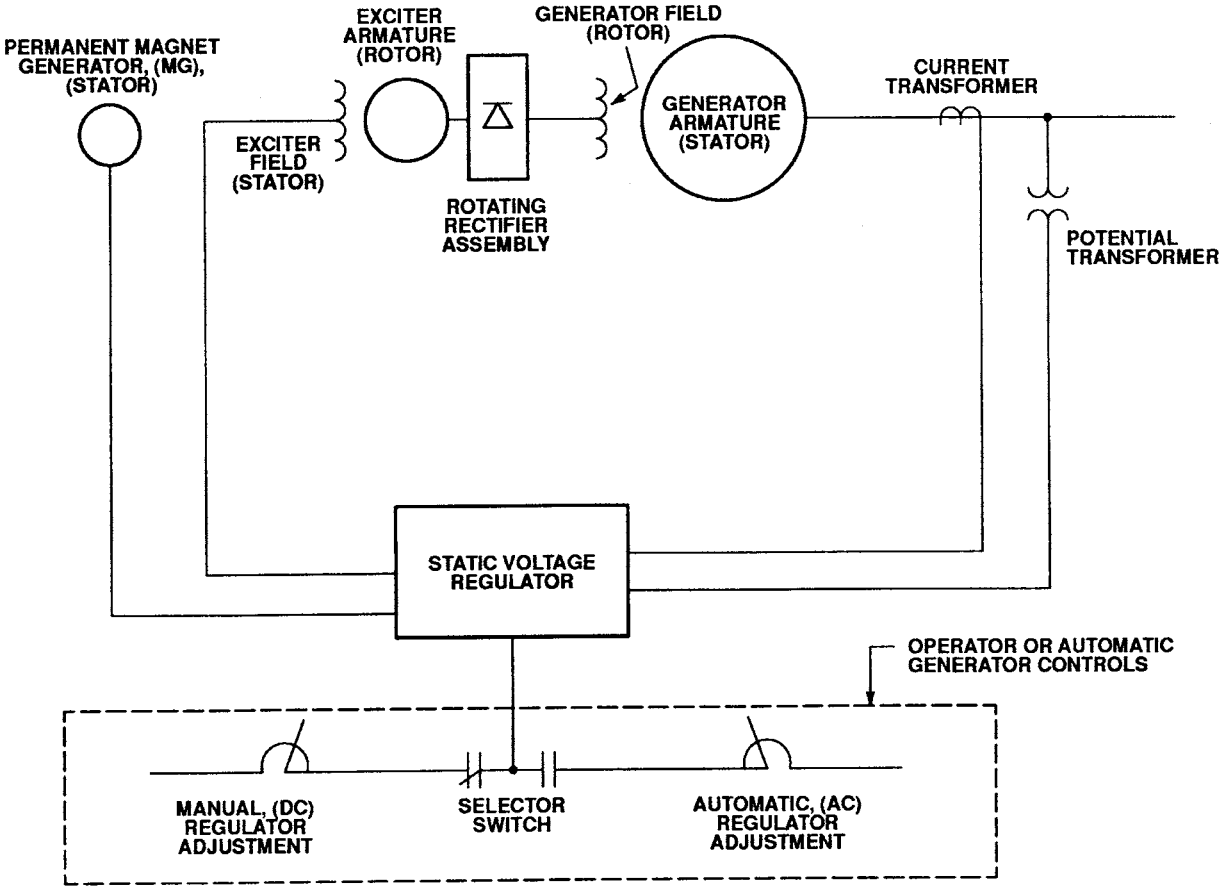


Figure 4. Brushless Exciter - Typical Layout

At this time there are a few Canadian GE digital exciters in service, designated Silco 5. It is projected that by mid 1993, exciters designated EX2000, (manufactured by GE in Salem, VA), will be finding their way to customer locations. The EX2000 will be available as either a Potential Source or Compound Source system. It is expected that the EX2000 will bring the cost of excitation systems down, due to the use of components that are common with other GE equipment, (i.e.--drive systems for motors). The primary difference between a digital exciter and analogexciter is the way one interfaces and makes changes. The digital exciters will require a PC, (with RS232 interface), to access, monitor, and make modifications. The EX2000 product requires use of a software tool-kit called ST2000 which provides a convenient way of accessing, modifying, and monitoring the process.

Prior to the start of any installation, adjustments, or checkout; it will be necessary to review drawings, instructions, and data for the unit. Proper preparation, with attention to detail, will help avoid making mistakes and can save considerable time later on.

Experience has shown that it is best to obtain one complete set of excitation instruction books, generator information, and generator data. These can be inserted in a binder, and used as a working reference. This can be fairly time consuming as it often requires using a jobsite copying machine, to reproduce a relatively large amount of material, taken from the turbine-generator instruction manuals.

Gas turbine-generator drawings can become a bit "challenging", because of the numerous interconnections and the numerous GE equipment suppliers, plus other vendor and customer drawings on top of all this. Unfortunately, these drawings are not "gathered up" under one set of elementary, interconnect, and connection drawings. It is easy to make a mistake, (or encounter difficulty), at a point where different parties "tie" their equipment together. Experience has shown it is best to obtain at least (2) complete sets of all drawings, .(3) sets are even better, before starting work. One set of drawings will be marked up, and used to prepare "as built" drawings; these will then be returned for revision. The second set will be marked up and retained at the site until revised drawings are issued, and if a third set is made they may be retained in the responsible GE office, (to handle any problems that may occur until revised drawings are issued). Failure to retain at least one extra set of marked up drawings, upon completion of the installation, can be "risky business". In addition to marking up the necessary GE drawings it usually becomes necessary to mark up customer, (and sometimes other vendor), drawings. Again, extra set of marked up drawings should be retained.

Typical GE drawings, (that would be required in performing the installation and checkout, relative to the exciter), would be; elementaries, connection, and interconnections for:

- Turbine Control Panel.
- Generator Control Panel.
- Generator Compartment.
- Generator Auxiliaries Compartment.
- Generator Excitation Compartment.
- Generator Terminal Enclosure.

These drawings will be found in the turbine-generator instruction manuals. Although the prime concern is the excitation system, the exciter controls and logic must interface with other equipment controls and logic; this then requires "straying away" from the exciter, into other areas, in order to be certain that the system functions properly. It is suggested the "working" drawings be copied and inserted into the "working reference binder", with the instruction manuals.

The exciter requires adjustments that are dependent upon the generator. This data must be located and reviewed, then, in a manner similar to that suggested for the instruction manuals and drawings; the data should be copied and inserted into the "working reference binder" with the instruction manuals and drawings. Typically, the following information will be required:

A. Generator Curves (from the generator instruction manual).

   1. Saturation curve.
   2. Vee curves.
   3. Capability curve.

B. Generator Data (from the generator instruction manual).

C. Customer Data (refer to Sheet A04 of the exciter drawings, and, refer to Sheet 2G of the generator control drawings. Both of these drawings will be located in the generator instruction manual).

D. Verify the:

1. Potential Transformer ratio (PTR).

2. Current Transformer ratio (CTR).

3. Generator field meter shunt ratio.

E. Make note of the: Generator field resistance, at a given temperature.

In order to perform a successful start-up, (with a minimum of problems, risk, and off line operation), it will be necessary to test, adjust and calibrate numerous items. If the excitation system instruction manual is followed, many of the adjustments will be made with the unit running. Running a turbine-generator to perform adjustments, that are easily made using test equipment, will be expensive, present a degree of risk, and delay placing the unit in operation. Also, choosing to not verify equipment calibration and operation, (with the thought that it is "factory tested and calibrated"), can be risky. Therefore, it is appropriate to perform as much "non-running" calibration and test work as is practical. Even if the equipment is found properly functioning and calibrated, the assurance that a dependable excitation system will be "going to work", can relieve some of the pressure associated with a new unit start-up. The drawback, in performing simulated tests. Using test equipment, will be the necessity of understanding the excitation system and its operation; there is no instruction book procedure to completely guide one thru such tests and calibration. The following test equipment is suggested, if it is not available it may be rented. In comparison to the cost of operating a turbine "off line", and the cost of delays in placing a unit in operation; test equipment rental or purchase is very reasonable.

Typical Test Equipment Required

- Three phase Variac. Nominal 115 VAC output at 2.5 AAC. Used to simulate PT signal.
- Power Supply. 0 to 500 VDC at 1.5 ADC. Used to simulate generator field voltage.
- Single Phase Variac. Nominal 115 VAC output at 2.5 AAC.
- Single Phase Transformer. 480/240-240/120 VAC, 0.33 KVA or greater.
- Power Supply. 0 to 100 mvdc. Used to simulate generator meter shunt output, (generator field current simulation)
- Three Digital Multimeters (DMM's), 4-1/2 digit. One meter must have frequency measuring capability. (i.e., Fluke Model 87 or 8060A).

- Three Phase Power Supply. Variable voltage and variable frequency, nominal 120 VAC, 60 Hz, 2 AAC. (i.e., Pacific Power Source Model 308K-VR). May be used to simulate generator PT's in place of a Three Phase Variac.

- Two Power Supplies. 0 to 7 VDC, 100 ma, requires accuracy to approximately 1 mv. Used to calibrate generator field temperature sensor. Not required for Potential Source or Shunt SCR systems.

- Oscilloscope. Dual channel, 20 MHZ or better. With (2) probes capable of safely measuring 600 VDC, (usually 100X probes). Also, (2) probes for low voltage signals, (i.e.-10X probes). It is highly desirable to have a current probe available to look at SCR gate signals on Potential Source and Shunt SCR systems, (i.e.-Tektronix P6021).

- Five Light Sockets and Bulbs. Common 120 VAC porcelain bases or "pigtails", with 100 or 150 watt bulbs. Used to simulate the generator field, in verifying the operation of Potential and Shunt SCR systems only.

- Three Phase Electrical Power. Nominal 15 AAC, 208 or 240 VAC. Used to provide power to 3 phase variac and power in checking the operation of Potential and Shunt SCR systems.

In lieu of a 0 to 500 VDC Power Supply, a full wave bridge rectifier, (rated 1000 PIV and 2.5 ADC), may be used, (with the already noted single phase variac and single phase, 480/240 - 240/120 VAC, 0.33 KVA transformer), to obtain a source of 0 to 500 VDC.

The following equipment is required to verify the calibration and operation of the Underexcited Reactive Ampere Limit, (URAL), circuit, without operating the unit on line. Utility companies normally furnish this equipment to their protective relay technicians.

- Phase Shifting Transformer. 250 VA or greater. (i.e. - States/Multiamp).

- Phase Angle Meter. (i.e. - Dranetz phase angle meter).

- Load Box. (i.e.- States/Multiamp or TESCO).

- AC Ammeter and Voltmeter. (i.e. - Dranetz Polymeter).

- Three Phase Variac. (Previously p1d in the list).

It may be found that there is other equipment available, that will be more suitable, in calibration of the URAL circuit. (i.e. the more up to date electronic relay test sets, by Doble or Multiamp, that have the capability of producing a phase shifted current and voltage signal).

The EX2000 exciters will also require a PC, with RS232 interface, and ST2000 software tool-kit. In addition a high speed recorder such as the Astro-Med DASH IV or Hioki 8830.

There is a certain "intimidation" period that seems to go with the arrival at a new jobsite. Once associates are met, drawings, data, and instruction books are reviewed; and a physical inspection is made, this period is ended. Plans are made and checkout can begin. Usually, a Field Engineer experienced in excitation work is brought in to perform pre-start-up checks, adjustments, and to be on site for start-up. Normally the equipment will be installed, and the wiring completed, before this individual arrives. It will be necessary for the Field Engineer to procure all drawings, data, instructions, and test equipment; if this has not previously been done. For this reason, it is best to discuss these requirements, with the Field Engineer, in advance of their arrival at the jobsite. This will shorten the "on site" preparation period, allowing work to begin sooner.

It is necessary to verify connection and interconnection wiring of the controls and logic circuits. In addition, the circuits themselves must be verified for proper operation. This can sometimes be performed in a combined check. The checks will normally be done by the Field Engineer, with the co-operation of any other interested parties, (i. e. Turbine Controls Specialist, user customer, architects-engineers, etc.). The equipment will extend beyond the excitation system, reaching into other control systems that may be furnished by other GE components or other vendors. With good co-operation, between interested parties, it may be possible to verify several circuits simultaneously, avoiding repeat checks. It must be remembered that this will be "untried and unproven" equipment, going into service. Any circuit that is not verified cannot be trusted.

The Field Engineer will normally perform adjustments and checks of the various controls, prior to start-up. Settings should be recorded on data sheets with the data retained for both the customer and the responsible GE office. This information will be used during future maintenance checks. The ability to perform this work will depend entirely upon having various pieces of test equipment available.

If thorough pre-start-up checks and adjustments are completed, very little will have to be done during start-up. There are a number of items that will have to be attended to, but pre-start-up checks will eliminate most of this work. On the other hand, if pre-start-up checks have not been performed and these checks and adjustments are to be made with the turbine running; plan on several hours of "off line", (no load), generator operation. Once again, adjustment data should be recorded. It will also be necessary to perform "on line" URAL calibration checks, if this

(optional) circuitry was not tested during pre-start-up checks. Numerous problems and delays will typically occur, during a start-up; and for this reason it is not unusual to work long hours with little or no rest. Long hours, (without sleep), unexpected problems requiring immediate attention, pressure from other parties to complete work quickly; all of this while making adjustments on operating energized equipment simply put: do pre-start-up adjustments. Different excitation systems have their own particular concerns, but they all have in common the start-up verification of certain operational points. Some nominal period of "off line" operation time must be allocated for these checks.

The excitation system is only one of the many systems that may be of concern. Often times the Field Engineer is called upon to perform checks, adjustments, and offer start-up assistance to other systems. If other systems are also to be included, then adequate time and assistance will be required. Also, as with the excitation system, drawings, instructions, and data; along with test equipment, will be needed.

Typically, protective relays are included in the generator controls. In addition the customer may include other protective relaying, located separate from the gas turbine-generator. It is absolutely essential that these devices function properly. The protective relays should be verified, as to their connections and interconnections, functional operation, (that is to say they will trip the required circuit breaker), and calibration. This work can be quite time consuming, requires specialized equipment, and specialized knowledge. It is usually best to "farm out" the relay testing, to people who specialize in this work, even though the Field Engineer may be able to perform this duty. Prior to relay calibration, relay settings must be obtained from the power system engineering group, responsible for system protection. This may be GE, the user customer, or an architect-engineering firm.

The generator controls are usually verified by the Field Engineer. They will include metering, and with a few exceptions, it is usually more convenient to include the metering with the protective relay calibration and testing.

Often a main generator breaker is included as part of the generator package. It will be necessary to verify this breaker's operation and adjustments. Should the switchgear encompass a line-up of several breakers, then plans must be made for their checkout. It is not likely that time will permit the Field Engineer, (who may be qualified to direct this work), will be able to follow thru on the checkout of more than one circuit breaker.

The generator may receive pre-start-up tests and checks by a Generator Specialist. Or, it may be up to the Field Engineer to follow thru on it's readiness for service. Data sheets should be prepared for the various system tests; as work progresses it may be necessary to repeat some tests.

In the process of connecting the generator, bus work and cables will suddenly become of concern. Experience has shown that it is best to keep track of this work; this may require that the Field Engineer divide his attention between excitation and controls work and the generator connections. On some rare occasions, knowledgeable, caring craftsmen will be assigned to make the generator connections. In this latter case, keeping track of the work is greatly simplified. GE information on bus connections, cable terminations, and insulation, leaves room for improvement.

During the installation of a turbine-generator, systems reach a stage where they will be made operational and tested. During these tests it is often necessary to by-pass protective devices, communications is often poor, controls are not within site of the equipment, and craftsmen will unknowingly enter the vicinity of equipment that is about to be powered up. In the case of the exciter, the greatest areas of concern have to do with energized equipment. During exciter tests, unless precautions are taken, it would be possible to inadvertently power up:

- Generator brush rigging.
- Potential transformers, (PT's).
- Power potential transformers, (PPT's).
- AC and DC control circuits.
- Power bridge rectifiers.
- Bus and cables.
- Fans and heaters.
- Circuit breakers.
- And perhaps other items.

Warning signs and barrier tape should be utilized, where needed. In addition, it is advised to stop, think., and discuss the test with associates; before "powering up". As is possible, try to set up a method of communications with remote locations; both a dedicated radio and telephone are highly desirable, but don't expect to be provided either one.

In addition to the personnel safety hazard, there is also the hazard of damaging equipment. This means that planning must be made before each system test. Once again stop, think., and discuss the test before "going to work".

A Lifted Lead & Jumper Check List should be prepared, and placed in a conspicuous location. The check list will help ensure protection and control circuits are restored, once testing is completed. To help minimize the possibility of equipment damage, (in control circuits), it is a good idea to use jumpers that are fused. Jumpers that incorporate fuses can be easily constructed and the fuse ratings can be adjusted to meet differing requirements, (a 2 ampere fuse works quite well, in most instances).

Excitation work can present some unique challenges and opportunities. Devising test and check methods, (and equipment), that will ensure equipment start-up with maximum reliability, minimum risk, and reduced delay are required. Along the way there will be opportunity to practice patience, diplomacy, forgiveness, and originality.